This Zenodo deposit contains a publicly available description of the Dataset:

"Single nuclei RNA sequencing of postmortem cingulate cortex,midbrain and motor cortex of healthy donors and Parkinson's disease patients – 10x multiome (snRNA-seq and snATAC-seq).".

Dataset Description:

This dataset consists of raw sequencing snRNA-seq data and snATAC-seq data (10x Genomics Chromium Next GEM Multiome ATAC/GEX). The data is part of an overall set of samples derived from postmortem midbrain (n=140), cingulate cortex (n=190) and motor cortex (n=4) of healthy donors (n=114), patients with Parkinson's disease (n=75) or patients with other neurological disorder (n=1). The protocol followed to isolate nuclei from postmortem brain samples and to prepare sequencing libraries can be found here: https://www.protocols.io/view/nuclei-isolation-and-permeabilisation-of-fresh-fro-n2bvj3yqnlk5/v1, https://cdn.10xgenomics.com/image/upload/v1728078404/support-documents/CG000338_ChromiumNextGEM_Multiome_ATAC_GEX_User_Guide_RevG.pdf.pdf. To increase throughput and to decrease batch effects, several donors have been pooled together into a single sequencing library. To computationally demultiplex the nuclei to their corresponding donors, cellsnp-lite (version commit: aad18644adcde853c313362a856a24245c9b91f7) followed by vireo (https://github.com/single-cell-genetics/vireo/pull/108) has been used. The population VCF with the donor genotypes derived from whole genome sequencing data has been used to assign nuclei back to their donors.

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Dataset Name: voet-pmdbs-sn-multimodal, v1.0

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ASAP Project: Understanding inherited and acquired genetic variation in Parkinson's disease through single-cell multi-omics analyses: a unique data resource

Project Description: The functional roles of inherited and acquired genetic variation in the pathogenesis of Parkinson's disease (PD) remain largely unknown. Here, we will first study how germline genetic variants at PD-risk loci, which were identified by genome-wide association study (GWAS), perturb the expression of genes in specific cell (sub)populations of the brain and gut. To this aim, we will apply single-cell gene-expression and open-chromatin quantitative trait locus (QTL) analyses, enabling identification of PD-relevant genes and cell (sub)types. We will deliver the mechanisms of PD-candidate gene expression (dys)regulation in the normal condition, with ageing and in PD, as well as a unique single-cell multi-omic resource for the community. Second, we will study the nature and role of somatic mutations in brain and gut cells in PD- etiopathology. Finally, we will characterize biochemical and phenotypic effects of loss- or gain-of-function QTLs and somatic mutations of candidate genes in in vitro and in vivo model systems, including their impact on the neuro-immune axis.

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